

Tech stack

Frontend

Next.js + TypeScript
Tailwind CSS

Backend

Next.js API routes / Server Actions

Database

Supabase PostgreSQL

Authentication

Supabase Auth

Storage

Supabase Storage

Data Processing

TypeScript initially
pandas/Python later if needed

AI

LLM API + structured outputs

Validation

Zod

Charts

Recharts

Deployment

Vercel

NFC

NFC tag → Quick Expense URL

Core Workflow

1. Bank / Wallet Statement Upload

BDO / GCash / Credit Card statements



Bulk Upload

2. Transaction Extraction

Extract transactions from uploaded statements and convert them into a standardized transaction format.



3. Duplicate Detection

Check imported transactions against existing records using predefined duplicate-detection rules.



4. Transaction Classification

Determine whether each transaction is:

- Internal Transfer
- Income
- Expense
- Unmatched / External Transfer



5. Internal Transfer Reconciliation

Match transactions across accounts to identify transfers between the user's own accounts.

Example:

BDO: -₱5,015

GCash: +₱5,000

- Internal Transfer: ₱5,000
- Transfer Fee: ₱15
- Expense: ₱15

Internal transfers are excluded from expense and income totals so that moving money between the user's own accounts does not inflate financial activity.

↓

6. AI Expense Categorization

Actual expenses are categorized using AI based on the user's configurable category list.

Example:

Jollibee → Food
Grab → Transportation
Netflix → Subscriptions

↓

7. Confidence Evaluation

The system evaluates the confidence of the AI categorization.

- High confidence → Automatically categorized
- Low / uncertain confidence → Requires user action

The user is not required to review confidently categorized transactions.

↓

8. User Review

For uncertain transactions, the user selects or confirms the appropriate category.

User corrections can be retained as rules/preferences to improve future categorization.

↓

9. Trusted Transaction Dataset

Processed transactions are stored as structured financial records containing:

- Date
- Amount
- Account
- Transaction type

- Merchant / description
- Category
- Source
- Classification status
- AI confidence
- Related transaction, where applicable

↓

10. Financial Dashboard

The processed dataset powers the dashboard, including:

- Income
- Expenses
- Net income
- Spending by category
- Spending over time
- Income vs. expenses
- Category trends
- Optional budgets

Internal transfers are excluded from spending totals and expense visualizations.

1. Account System

The user can create accounts such as:

BDO Savings
GCash
Cash
Credit Card
Other Bank

And:

+ Add Account

Each account has:

- Account name

- Account type
- Institution/provider
- Active/inactive status
- Import source/configuration

For MVP, I wouldn't worry about tracking balances.

The account exists primarily to establish **where a transaction came from or went to**.

2. Transaction Model

This is the most important part of the entire application.

Every imported or manually created transaction eventually becomes a standardized internal record.

Conceptually:

Transaction

- Date
- Amount
- Direction
- Account
- Merchant / Description
- Transaction Type
- Category
- Source
- Status
- Confidence
- Related Transaction

The system should distinguish:

Expense

GCash

-₱350

Jollibee

Food

Income

BDO

+₱30,000
Salary
Income

Internal transfer

BDO
-₱5,000

GCash
+₱5,000

Transfer fee

BDO
-₱15
Transfer Fee
Expense

The important thing is that **the ₱5,000 itself does not become an expense.**

3. Transfer Reconciliation

This is one of the more interesting parts of the project.

Given:

BDO
-₱5,015

GCash
+₱5,000

the system tries to find the corresponding transaction across your other imported accounts.

It identifies:

BDO → GCash
Transfer amount = ₱5,000
Fee = ₱15

The final financial dataset effectively becomes:

Transfer: ₱5,000 → excluded from spending
Expense: ₱15 → included in spending

So your dashboard doesn't say:

"You spent ₱5,015"

It says:

You spent ₱15.

And importantly, the ₱5,000 isn't even treated as something that needs to appear as a major component of your expense charts.

4. Unmatched Transfers

Your answer to #4 is important.

Suppose:

BDO
-₱1,000

but there is no corresponding:

GCash
+₱1,000

in the imported data.

The system shouldn't automatically assume:

"This is an internal transfer."

Instead, it can classify it as something like:

Transaction Type:
Transfer / External Transfer

Status:
Unmatched

and therefore count it as an expense **unless/until you identify it otherwise.**

That gives us a useful principle:

Only transactions that can be reconciled as internal transfers are excluded from spending.

That protects the integrity of your analytics.

5. Bulk Import

The intended experience is basically:

Upload

[BDO September.csv]

[GCash September.csv]

[BDO Credit Card September.csv]



PROCESS FILES



1,248 transactions



1,221 categorized

20 uncertain

7 duplicates



DONE

No manually entering 1,248 transactions.

And you don't need to worry about the actual CSV parsing architecture yet.

6. Duplicate Detection

You specifically want this governed by defined rules rather than simply:

"These two transactions look similar."

Good.

The system should have a deterministic duplicate-detection layer.

Conceptually:

Transaction A
+
Transaction B
↓
Duplicate Rules
↓
Same transaction?
↓
YES → suppress duplicate
NO → retain both

This is important because you'll potentially be uploading overlapping statements.

For example:

September 1–30

then later:

September 15–October 15

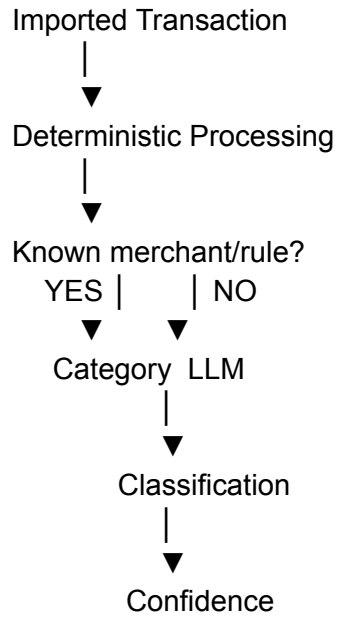
The overlapping transactions shouldn't double your spending.

7. AI Categorization

This is where the LLM comes in.

I wouldn't make the LLM responsible for everything.

Instead:



For example:

JOLLIBEE FO
₱285

→ Food
→ 98% confidence

No action required.

But:

XYZ ONLINE SERVICES
₱1,499

→ Shopping?
→ Electronics?
→ Subscription?

Confidence: 41%

That becomes an **action required** item.

8. Your AI Review Philosophy

This is probably the most important UX requirement you gave me:

Don't make me review everything. Only make me act when the system isn't sufficiently confident.

So the UI shouldn't be:

"Review 1,248 transactions."

It should be:

20 transactions need your attention.

Something like:

ACTION REQUIRED

20 transactions need categorization

₱1,499 XYZ ONLINE SERVICES

Suggested: Shopping

Confidence: 42%

[Shopping]

[Subscription]

[Other]

Once you select one, it's done.

And your correction can become a rule/preference for future transactions.

9. Categories

Keep the default taxonomy relatively small.

Something like:

Food

Transportation

Shopping

Bills

Entertainment

Health

Education
Subscriptions
Personal
Other

But:

You can add/edit categories.

And AI works against **your category set**, not some fixed OpenAI category taxonomy.

That's important.

If you create:

Judo

then the AI can classify:

Judo Philippines → Judo

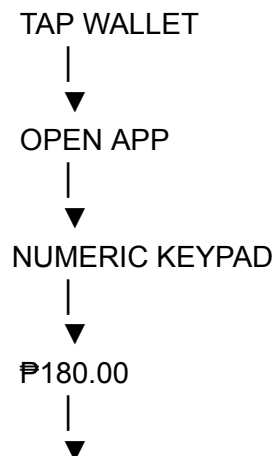
rather than forcing it into:

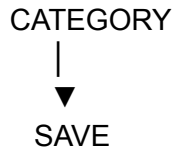
Sports / Entertainment / Other.

10. Cash / NFC

The NFC feature should be extremely simple.

Your desired flow:





That's it.

No merchant.

No description.

No giant form.

You can optionally add a note later, but **don't make it part of the required flow**.

The NFC itself can come later. The underlying:

Quick Cash Expense

feature should exist independently first.

11. Income

You said income tracking is required, but you don't necessarily need sophisticated income classification yet.

So the system should distinguish:

INCOME
+₱30,000

from:

TRANSFER
+₱5,000

and:

EXPENSE
-₱500

But we're **not** building a huge income-management system.

That's enough for:

September

Income ₱35,000
Expenses ₱21,450
Net ₱13,550

with internal transfers excluded.

12. Budgeting

Optional feature:

Budget

Food
₱5,000 / month

Transportation
₱3,000 / month

Entertainment
₱2,000 / month

If you don't configure a budget, nothing happens.

If you do:

Food
₱4,320 / ₱5,000

This should **not** become a major MVP feature.

It's basically:

category + monthly spending limit + progress.

13. Dashboard

You want all the major views except account balances.

I'd make the MVP dashboard:

Summary

SEPTEMBER 2026

Income
₱35,000

Expenses
₱21,450

Net
₱13,550

Expense breakdown

Food	₱6,200
Shopping	₱4,500
Bills	₱4,000
Transportation	₱3,100
Entertainment	₱2,100
Other	₱1,550

Spending over time

Line/bar chart showing daily or weekly expenses.

Income vs expenses

Income	
Expenses	

Category trends

Potentially:

Food	
August	₱4,200
September	₱6,200

This lets you answer:

"Where is my money actually going?"

without needing a complicated finance app.

14. What I would NOT put in MVP

This is important because your idea could get bloated very quickly.

Not MVP:

- ✗ AI financial advisor
- ✗ Investment tracking
- ✗ Net worth tracking
- ✗ Account balance synchronization
- ✗ Bank API integrations
- ✗ Automatic bank login
- ✗ Receipt OCR
- ✗ Forecasting
- ✗ Complex budgeting system
- ✗ Shared/family accounts
- ✗ Multi-user functionality
- ✗ Mobile app
- ✗ Fancy financial reports

MVP:

- ✓ Multiple accounts
- ✓ BDO import
- ✓ GCash import
- ✓ Credit-card import
- ✓ Bulk uploads
- ✓ Normalize transactions
- ✓ Duplicate detection
- ✓ Internal transfer reconciliation
- ✓ Transfer fee handling
- ✓ Income vs expense distinction
- ✓ AI categorization
- ✓ Confidence-based review
- ✓ Custom categories
- ✓ Correction/learning rules
- ✓ Cash expense entry
- ✓ Basic NFC launch flow later
- ✓ Optional budgets
- ✓ Dashboard + charts

